

FLAT-PACK BENCH: Evaluating Spatio-Temporal Understanding in Large Vision-Language Models through Furniture Assembly

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[flat-pack-bench.github.io](https://github.com/flat-pack-bench)

Abstract

The emergence of Large Vision-Language Models (LVLMs) has significantly advanced video understanding capabilities. However, existing benchmarks focus predominantly on coarse-grained tasks such as action segmentation, classification, captioning, and retrieval. Furthermore, these benchmarks often rely on entities that can be easily identified verbally, like household objects, animals, human subjects, etc., limiting their applicability to complex, in-the-wild video scenarios. But, many applications such as furniture assembly, cooking, etc., require step-by-step fine-grained spatio-temporal understanding of the video, which is not sufficiently evaluated in current benchmarks. To address this gap, we introduce FLAT-PACK BENCH, a novel benchmark centered on furniture assembly tasks. Our benchmark evaluates LVLMs on nuanced tasks, including temporal ordering of assembly actions, temporal localization of assembly state, understanding part mating, and tracking, using multiple-choice questions paired with visual prompts highlighting relevant parts as references for fine-grained questions. Our experiments reveal that state-of-the-art LVLMs struggle significantly with fine-grained spatio-temporal reasoning, highlighting their limitations in effectively leveraging temporal information from videos, limited tracking ability, and understanding of spatial interactions like physical contact.

1. Introduction

Imagine an AI assistant that helps us out with complex but practical everyday tasks, like cooking a complicated recipe, repairing equipment, or assembling furniture. We may want this AI assistant to watch an instructional video or demonstration of the task, then answer questions to help us out: what ingredients should I add next? Which pipes should connect? Which of these pieces should be screwed in first? A natural way to build such an AI assistant is to use Large

Vision-Language Models (LVLMs) [1, 10, 19, 28, 41, 42, 48] – after all, these models exhibit broad understanding of images and videos and can interact in natural language. But are current LVLMs up to the task?

To answer this question, it is useful to walk through the required skills. The first step of course is understanding the demonstration video. This itself presents significant challenges. Simply understanding the overall objective (“What is being done in this video?”) is not enough: the LVLM must understand what each individual step is, how to accomplish that step, and when to execute it, and then communicate this understanding with a human user. Understanding how to execute individual steps requires the model to detect and localize particular objects (e.g., recipe ingredients or furniture parts) and recognize how they interact (e.g., detect when two parts are screwed together). Understanding when to execute what step requires tracking the ingredients or assembly components over long demonstrations with multiple steps. Finally, communicating this understanding in cluttered environments requires the model to understand not just text but also spatial references [39, 46].

Existing benchmarks do not stress-test LVLMs on such challenges. Many video QA benchmarks focus on short videos [11, 34], or on coarse-grained questions about video themes that do not require temporal understanding or tracking [17]. Often the scenes are uncluttered [4, 32, 44] making object references unambiguous [5, 39, 46]. There is also limited focus on modeling object interactions that are ubiquitous in complex, everyday activities.

In this paper, we address this gap with a new video QA benchmark using the task of furniture assembly as a sandbox (Figure 1). Furniture assembly is a simplified microcosm of precisely the challenges we identify above: recognizing and tracking objects over multiple steps and detecting their interactions in a cluttered visual scene. The simplicity of the domain (rigid parts that retain shape and identity throughout the video) allows us to precisely articulate the skills that current models lack. Success in this domain is a prerequisite for more complex domains where object state might

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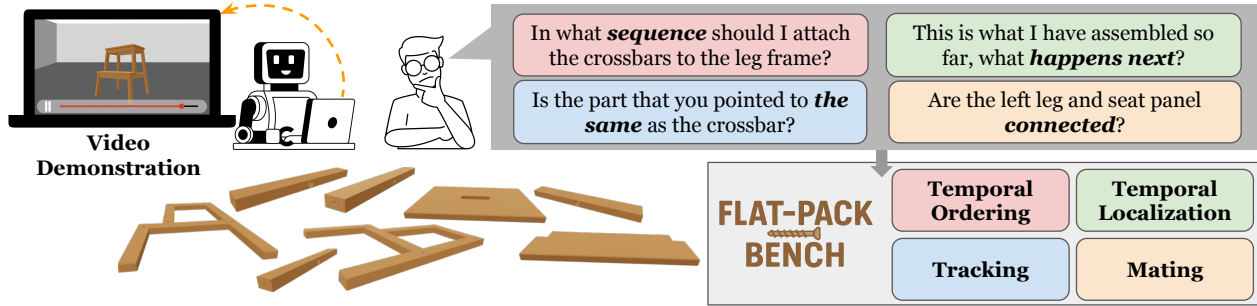


Figure 1. **Motivation for FLAT-PACK BENCH.** For AI assistants to understand an assembly process through observation, they need to be adept at fine-grained spatio-temporal reasoning about the video. We propose FLAT-PACK BENCH to evaluate Large Vision-Language Models on four such fine-grained video understanding tasks, namely – Temporal Ordering, Temporal Localization, Tracking, and Mating.

change (e.g., tomatoes getting cut or squished) and complex interactions between ingredients and parts might occur.

To build this benchmark, we augment prior furniture assembly datasets [15] with significant new annotations: segmentations of each part for spatial references, connections between individual parts, and a manually curated set of natural language multiple-choice questions. The resulting benchmark, FLAT-PACK BENCH¹, tests multiple axes of temporal understanding, including whether models can track parts across frames, understand which parts connect to which, and determine the order in which different connections happen.

We test multiple proprietary and open LVLMs on our benchmark and find that even the best models struggle: OpenAI’s latest GPT-5 model [19] achieves an accuracy of $\sim 38\%$, trailing far behind human performance of 94.18%. We look deeper into model performance, and show how models struggle with spatio-temporal reasoning tasks like tracking and contact detection required for the tasks in our benchmark. We further explore an agentic approach [25] that uses standard, state-of-the-art vision models like SAM2 [23] as tools, but find that the tools themselves struggle on these challenging videos. Our benchmark suggests that, despite rapid progress, current LVLMs (and computer vision systems in general) have limited capability to understand the temporal evolution of complex scenes.

2. Related Work

Video Understanding Benchmarks. Prior works on Video Question Answering (VidQA) tend to focus on coarse-grained questions about high-level scene semantics [4, 8, 11]. Recently there has been a surge of interest in benchmarks on physical scene understanding [32, 44] but these often focus on synthetic videos without clutter or occlusion. On more real-world videos, some recent benchmarks seek to evaluate spatial intelligence [37], but much of this work focuses on static scenes with camera motion (typically ignoring any dynamic objects) [12, 37, 40, 45]. VLM4D [47] evaluates

the relative motion understanding of LVLMs for dynamic scenes, but unlike our proposed benchmark, does not explore interactions between objects. Temporal understanding abilities of LVLMs has also been studied with past works evaluating temporal sensitivity [35] and eliminating single-frame bias [49] in existing benchmarks. Our benchmark is also related to these works, but we focus on the fine-grained temporal evolution of the scene. Interactive video-based guidance benchmarks [2, 3, 20, 30] are also related, but they tend to focus on short-range temporal context: what has immediately transpired and what should happen next.

Similar to our approach, LEGO-Puzzles evaluates LVLMs on multi-step assembly-based reasoning [26]. However, they focus on a *multi-image* setting, providing 2-3 images of relevant assembly steps as inputs to the model. This simplifies the problem and is not representative of real-world demonstrations which may have long demonstrations with multiple frames, as represented by our benchmark: the model must decide what frames it should focus on. Finally, recent benchmarks have also explored long videos and fine-grained questions [17, 33], but often they focus on simple uncluttered scenes. Cluttered scenes, like those in our benchmark, introduce additional challenges where the model has to disambiguate similar-looking parts, and understand not just textual but spatial references.

Regional Understanding in LVLMs. Recent work has explored tasks where the LVM must track a segmented object through a video and reason about the track [39, 46]. However, the objects in these datasets are small in number (typically one or two per video) and easily tracked. In contrast, our benchmark features manually curated and verified segmentations and questions involving complex interactions among multiple, visually similar parts—making it substantially more challenging, as it requires more difficult tracking. Another related model–benchmark pair, PerceptionLM and PLM-VideoBench [5], assume tracking to be solved, with the model provided full-video tracks, whereas our benchmark requires the model to track the highlighted regions specified in specific frames. A related task is Spatio-Temporal Visual Grounding (STVG) [18, 24, 27, 43], where models are given

¹Flat-pack furniture is a term used for ready-to-assemble furniture. Benches are commonly sold in flat-pack form, thus: FLAT-PACK BENCH.

Table 1. **Dataset composition:** Shows the number of videos (#V), questions (#Q), and templates per category (#T), along with average questions per video (Q/V), per template (Q/T), and unique templates per video (uT/V).

Category	#V	#Q	Q(%)	Q/V	#T	Q/T	uT/V
TLOC	35	103	17.11	2.94	6	17.17	1.71
TORD	39	155	25.75	3.97	2	77.50	1.51
MATE	21	87	14.45	4.14	3	29.00	1.48
TRACK	43	257	42.69	5.98	2	128.50	1.37
Total	50	602	100.00	12.04	13	46.31	4.18

a video and a textual description of an object, and the model must segment the referred object. In contrast to this problem, we do not expect segmentation masks as an output, though we do demonstrate the performance of an agent that uses segmentation/tracking as a tool.

3. FLAT-PACK BENCH

We now describe the process of creating the benchmark.

Data. We base our benchmark on the IKEA-Manuals-at-Work (IMaW) dataset [15]. IMaW contains in-the-wild videos of people assembling IKEA furniture pieces. The dataset also comes with a limited set of annotations: 3D models for the furniture and the furniture parts, and key frames that are annotated with segmentation masks, the 6DoF pose for each part, and the *sub-assemblies* (a set of parts) that are being connected. The original assembly demonstration videos contain unnecessary segments consisting of text-only instruction cards, typically when the timestamps of consecutive key frames have a gap greater than one second. We use this heuristic to trim such sections from the raw videos, resulting in the *trimmed videos*. Together with videos composed solely of key frames (*key-frame videos*), these form the two types of assembly videos used in our evaluations. Key-frame videos contain only the most salient frames, offering a concise view of the assembly process, but require manual curation and are therefore unrealistic. Trimmed videos are more realistic, but often longer, stressing current model capabilities. We evaluate both settings in our experiments.

Visual Prompts. One challenge with constructing questions on furniture-assembly videos is unambiguously referring to furniture parts. Text descriptions (like “top railing”) may be ambiguous in symmetric structures, and may prompt the model to rely on its common-sense knowledge of typical furniture rather than the provided visual input. Instead, we propose to use *visual prompts* [36], and mark out each part on an image prompt using a label and a segmentation. Thus, each question in our benchmark consists of an assembly video, 1-2 visual prompts (constructed from frames in the same video) and a multiple-choice question. We use the

part’s label in the visual prompt to refer it in the question.

Visual prompts require segmentations. However, we find the segmentation annotations in IMaW to be incomplete (See App. A.2 for details). We therefore augment IMaW with manually annotated part segmentations on 343 frames from a subset of 50 videos. These frames serve as our visual prompts. Next, we construct the questions in the dataset.

Question Types. To fully understand a furniture assembly, the model must identify which parts connect (*mating*) and when they connect. To be able to infer which connections happen when, it should *track* the individual parts through the video. Thus, we propose the following question types:

- MATE asks the model to determine whether two parts are connected in the final assembly.
- TRACK provides two segmented frames with shuffled part IDs as visual prompts and requires the model to recover the correct correspondences using the video.
- TORD evaluates whether the model can infer the correct order of connection events.
- TLOC tests its ability to identify events immediately before or after the state shown in the visual prompt, assessing temporal localization and reasoning about nearby events.

For each of these question types, we create templates; some examples are shown in Fig. 2. Formulating these questions requires annotations of part–part connections in the video. However, as noted earlier, IMaW provides only sub-assembly–level annotations. As such, we augment each video with fine-grained part assembly annotations that specify which parts connect, to which other parts, and when.

Manual Question Curation. With these annotations and question templates, we can auto-generate questions. However, we found that auto-generation frequently produced questions that could be answered by ignoring the video and exploiting shortcuts. For example, auto-generated *mating* questions about parts already positioned for connection, or included distractor options with clearly distinct shapes or colors, enabling easy elimination (See App. A.3 for examples). To address this, we curated all questions manually using fixed templates. Annotators were provided the full assembly video, segmentation-labeled frames for visual prompts, the question templates, and detailed guidelines for avoiding shortcuts based on static cues from the visual prompt. The full list of templates is provided in App. A.3.

Final Benchmark. The final benchmark consists of 602 multiple-choice questions covering 50 different furniture assembly videos sourced from the IMaW [15] dataset. Figure 2 shows a snapshot and Tab. 1 shows the dataset composition. For additional details about the benchmark, see App. A.1. More examples from the benchmark can be viewed at: flat-pack-bench.github.io/viewer/

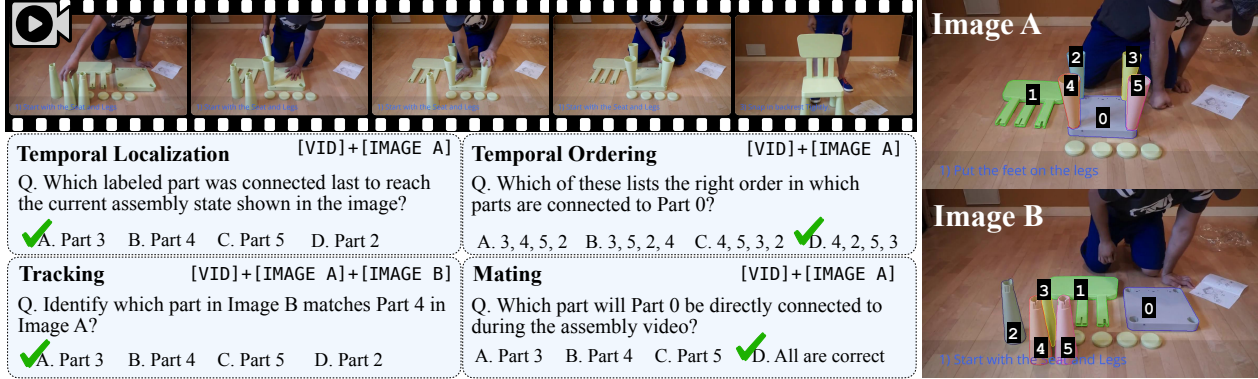


Figure 2. **Snapshot of FLAT-PACK BENCH.** Each question consists of an assembly video (top row), one or two visual prompts (Images A, B), and a multiple-choice question. The corresponding visual inputs are shown within each question box. Videos are sourced from the internet and may include artifacts like overlaid text. For clarity, part labels are enlarged, as the visual prompts are shown at reduced scale.

4. Evaluation on FLAT-PACK BENCH

4.1. Setup

Benchmark Models. We evaluated the following models:

- **Proprietary:** Gemini 2.5/3.1 [28] & GPT-5 [19].
- **Open:** Video-LLaVA [13], LLaVA-NeXT-Vid [41], LLaVA-OneVision [10], LLaVA-Video [42], Qwen 2.5/Qwen 3-VL [1, 22] and InternVL3 [48]. We cover models with a range of design choices, from fine-tuning image-text models with video data [41], to special architectural modifications to handle videos [1, 10, 48].
- **Specialized:** We also evaluate models tailored for specific capabilities relevant to our task: ArrowRL [35] improves temporal sensitivity; PerceptionLM [5] and VideoRefer [39] target fine-grained regional understanding; and GenS [38] selects question-relevant frames in long videos for a base LVLM (Gemini 2.5 Pro in our case).

We evaluate all the models under a zero-shot setting. Following VSI-BENCH [37], we use greedy decoding for all models (except for GPT-5, which does not support it) to ensure reproducibility.

Prompt Construction. The input prompt consists of a video, 1-2 visual prompts, the question, and a fixed task instruction that describes the multi-modal input and output format to the model. We employ three different settings to supply the visual prompt to the model:

- **Mixed-media Prompt:** The visual prompt is supplied as an image separate from the assembly video.
- **Collage Prompt:** Each frame of the video is a grid of images containing the visual prompt(s) (fixed in each frame) and the original video frame.
- **Concat Prompt:** The visual prompts are concatenated to the video as the first 1 (or 2) frame(s).

The task instructions describe the prompt format to the model. Previous works [38] have also shown that for long videos, finding salient key-frames for answering the questions can be challenging. To evaluate the effect of key-

frames, we also try both trimmed and key-frame videos for all prompt formats. We evaluate all configurations across all models, except when restricted by architectural or cost constraints. See App. B for more experimental details.

Chance Baselines. We also report the accuracy of choosing an answer uniformly at random (*Random Chance*) or picking the most common option per task (*Frequency Chance*).

Human Performance. We also evaluated how humans perform on the benchmark. We recruited a group of participants consisting of Computer Science students ranging from undergraduate to doctoral levels. Each participant was shown the assembly video, the visual prompt, the multiple-choice question, and the task instruction and asked to select an answer. We collected 3 responses for each question and selected the final response using majority voting. We also conducted a broader crowd-sourced study on a randomly-sampled subset of questions (see App. C for the details and results).

Metric. We use accuracy as our metric with a regex-based exact match to extract the answer from model responses.

4.2. Results

Table 2 shows our results. We show the score for the best-performing setting across all video types and visual prompt formats for each model (See App. C for more results).

Human Performance. As shown in Table 2, humans achieve very high accuracy (> 90% on all question categories), showing that spatio-temporal understanding is essentially second-nature to humans. 80% questions received unanimous responses, suggesting that our questions are clear and consistently understood (task-wise breakdowns in App. C).

Proprietary Models. Unlike humans, both GPT-5 [19] and Gemini 2.5/3.1 Pro [7] struggled on this task (37.71% and 33.72/32.89% respectively): only a modest improvement over the chance baseline (26.74%), and way below human levels. Choosing question-relevant frames using GenS [38]

Table 2. **Results on FLAT-PACK BENCH.** **Best model** and best open model are highlighted in each column.

Model	Rank	Micro Avg.	TORD	TLOC	TRACK	MATE
Human Performance	–	94.18	93.54	93.20	93.77	97.70
<i>Chance Baselines</i>						
Random Chance	–	26.41	25.00	25.00	25.49	33.33
Frequency Chance	–	26.74	27.74	30.10	26.46	36.78
<i>Proprietary Models</i>						
GPT-5	1	37.71	40.65	53.40	25.68	49.43
Gemini 2.5 Pro	2	33.72	40.65	44.66	23.35	39.08
Gemini 3.1 Pro	3	32.89	34.84	43.69	21.79	49.43
Gemini 2.5 Flash	4	31.06	31.61	41.75	23.35	40.23
Gemini 2.5 Pro + GenS	5	25.58	33.55	32.04	13.23	40.23
<i>Open Models</i>						
Video-LLaVA-7B	26	23.75	21.29	35.92	10.89	51.72
InternVL3-14B	5	37.71	42.58	21.36	37.74	48.28
InternVL3-38B	12	36.05	42.58	37.86	25.68	52.87
InternVL3-78B	1	41.03	43.87	39.81	42.02	34.48
Qwen2.5-VL-7B	22	30.23	27.10	18.45	33.07	41.38
Qwen2.5-VL-32B	13	35.88	34.84	29.13	33.07	54.02
Qwen2.5-VL-72B	2	40.37	41.29	30.10	45.14	36.78
Qwen3-VL-4B	11	36.54	34.19	33.01	32.68	56.32
Qwen3-VL-4B-Think	9	37.21	31.61	25.24	37.74	59.77
Qwen3-VL-8B	15	33.72	36.13	30.10	33.85	33.33
Qwen3-VL-8B-Think	17	31.73	34.19	33.01	25.29	44.83
Qwen3-VL-32B	6	37.71	38.71	46.60	31.91	42.53
Qwen3-VL-32B-Think	3	40.03	38.71	22.33	45.53	47.13
Qwen3-VL-30B-A3B	10	36.71	30.32	22.33	42.02	49.43
Qwen3-VL-235B-A22B	8	37.21	37.42	25.24	39.69	43.68
LLaVA-NeXT-Vid-7B	25	25.08	33.55	24.27	16.73	35.63
LLaVA-NeXT-Vid-34B	21	30.40	30.32	24.27	32.68	31.03
LlaVA-OneVision-7B	16	32.89	26.45	30.10	34.24	43.68
LlaVA-OneVision-72B	4	38.37	35.48	25.24	38.91	57.47
LLaVA-Video-7B	19	30.73	30.97	24.27	25.68	52.87
LLaVA-Video-72B	7	37.54	36.77	27.18	35.80	56.32
Perception-LM-1B	24	27.74	28.39	26.21	25.29	35.63
Perception-LM-3B	18	31.40	28.39	32.04	29.96	40.23
Perception-LM-8B	14	35.38	26.45	26.21	44.75	34.48
VideoRefer	23	28.57	32.90	30.10	17.51	51.72
ArrowRL-7B	20	30.56	30.97	24.27	29.18	41.38

did not improve Gemini 2.5 Pro’s performance. Thus, unlike these models’ success on previous video benchmarks, proprietary LVLMS struggled with the spatio-temporal understanding required to solve FLAT-PACK BENCH.

Open Models. Among open models, InternVL3 [48] (41.03%; 95% bootstrap CI [36.21, 45.64]) performed the best. Several open models, particularly the Qwen [1, 22] and InternVL3 [48] families, are competitive with proprietary models, but performance across open models is highly variable, with some only slightly above the chance baselines.

Specialized Models. Despite being trained specifically for fine-grained region-specific questions, PerceptionLM [5] and VideoRefer [39] perform poorly, likely due to a mismatch between their training data, which feature relatively simple videos with few fine-grained interactions, and the complex, multi-part interactions in FLAT-PACK BENCH. Still, PerceptionLM-8B is competitive with much larger models

(e.g., Qwen2.5-VL-32B), suggesting value in training on data rich in fine-grained interactions. Temporal sensitivity also helps ArrowRL [35] beat its base checkpoint (Qwen2.5-VL-7B), especially on temporal ordering (TORD). We also analyze correlations between model performance and factors such as video duration and difficulty (See App. C).

Taken together, these results indicate that current LVLMS—proprietary, open, and specialized—remain far from achieving the strong, reliable spatio-temporal understanding skills that humans demonstrate on FLAT-PACK BENCH.

5. Analysis

We now examine why LVLMS struggle on our benchmark. Unless stated otherwise, we use a standard open LVLM, Qwen2.5-VL-72B for analysis (See App. D for experimental details). Other open LVLMS, like InternVL3-78B, yield similar overall takeaways (App. E).

5.1. Linguistic Prompt Engineering

Previous work has shown that linguistic prompting strategies can improve model performance on language-based reasoning tasks [25, 29]. We evaluate whether such techniques also offer any benefit for the spatio-temporal reasoning required in our benchmark. We use key-frame videos for this experiment with Mixed-media prompts (as video type had little impact on accuracy, key-frame videos were faster to evaluate, and Mixed-media was the best prompt type for this LVLM; See Sec. 5.2). We consider the following approaches:

- **Zero-shot Chain-of-Thought Prompting [9] (ZS-CoT):** We prompt the LVLM to generate an explanation of their answer choice. We modify the task instructions to include “Please explain your answer step-by-step”. This baseline also uses greedy decoding.
- **Self-consistency w/. CoT [31] (SC-CoT):** We extend ZS-CoT with temperature sampling, generating 5 responses and then select the final answer with majority voting.

Table 3. **Effect of Linguistic Prompting Strategies.** Both ZS-CoT and SC-CoT fail to improve performance on FLAT-PACK BENCH.

	Micro Avg.	TORD	TLOC	TRACK	MATE
–	40.19	40.64	30.09	45.52	35.63
ZS-CoT	39.20	38.06	30.10	43.19	40.23
SC-CoT	32.23	32.90	24.27	35.02	32.18

Table 3 shows their performance, comparing them with vanilla prompts used in Sec. 4. Both approaches hurt performance, suggesting that unlike language-based reasoning tasks, CoT-based prompting does not improve spatio-temporal visual understanding required for our benchmark.

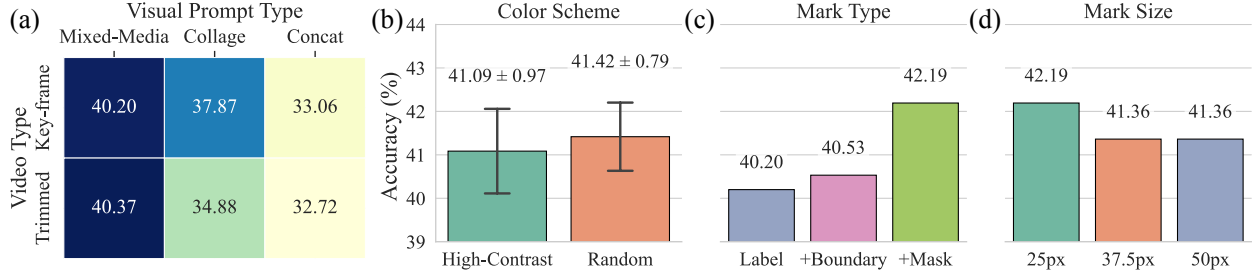


Figure 3. **Visual Data Ablation.** We study the effect of different strategies of providing the visual prompt and video processing (a). Next, we analyze how the (b) color scheme, (c) mark type, and (d) mark size affect the LVLM’s performance on our benchmark.

Table 4. **Image-only Prompt.** Performance of the LVLM using image-only prompts, along with the change (Δ) in performance from when the video is included in the prompt.

	Micro Avg.	TORD	TLOC	TRACK	MATE
<i>Video+Image</i>	40.20	40.65	30.10	45.53	35.63
<i>Image-only</i>	31.40	40.65	34.95	21.01	41.38
Δ	-8.80	0.00	+4.85	-24.51	+5.75
<i>Shuffled Parts</i>	29.96 ± 0.69	35.70 ± 3.55	36.25 ± 4.04	20.62 ± 1.68	39.85 ± 2.65
Δ	-10.24	-4.95	+6.15	-24.90	+4.21
<i>Human Perf.</i>	42.69	30.32	45.63	48.24	44.82

5.2. Effect of Visual Data Processing

Past work [14] has also shown that LVLMs are sensitive to visual data processing. We therefore evaluate the impact of the various choices around the visual input.

First, we look at the impact of trimmed vs. key-frame videos, and mixed-media vs. collage and concat prompts (Fig. 3(a)). We find that the video type does not have a big impact. However, among visual prompt types, mixed-media approaches perform the best, likely because of the mixed image and video training data for Qwen2.5-VL [1].

Next, we look at visual prompt rendering parameters. We use key-frame videos since the video type has little impact, and they are faster to evaluate. To create our visual prompts, we greedily sampled each mask color to be maximally distinct from the closest color already chosen and its underlying pixel to ensure visual distinctiveness. We also added a 2-pixel boundary around each mask to increase its saliency. We ablate through three axes:

- **Color Scheme:** Does greedily selecting a high-contrast color perform better than randomly picking a color? We render visual prompts across 3 runs and compare.
- **Mark Type:** We first overlay only the part IDs on the image, then the mask outline and finally, the mask itself.
- **Mark Size:** We gradually increase the marks’ font size.

We find that high-contrast colors (Fig. 3(b)) and mark size (Fig. 3(d)) have limited impact, but it is important to render all three: the label, the boundary and the mask (Fig. 3(c)).

5.3. Do LVLMs utilize temporal context effectively?

The high human performance on our benchmark shows that the questions can be easily solved using the provided infor-

mation, namely the assembly video and the visual prompt. Yet, most of the SOTA LVLMs struggle on our benchmark, particularly when the task requires careful temporal understanding, such as TRACK. The video is the main source of temporal information about the assembly in the question.

This leads us to ask the question: *Are LVLMs using the videos effectively?* To answer this question, we evaluated these models on an image-only version of our benchmark, i.e., the model was only provided the question text and the visual prompt image(s). We also measured human performance on this version (similar to Sec. 4). Table 4 shows the performance of the LVLM on this image-only version, and the change in their performance from the full evaluation, along with human performance. The sharp drop in human performance (>50%) shows that the questions do require videos to answer. We also observe that the overall performance of the model drops severely (8.80%), but mostly due to the TRACK sub-task. Accuracy on other tasks stays the same or improves, indicating that the LVLM *does not use the video effectively*, while humans use the video to answer.

Task-wise, performance improves on the MATE and TLOC sub-tasks. This suggests that the model used its superior image-understanding ability to recognize the part types (e.g., legs, backrest, etc.) and their positioning (e.g., hands holding/close to the next/previous part connected) in the prompt image, along with commonsense reasoning. Higher human performance on these sub-tasks, compared to TORD which requires long temporal context, also suggests that such image- and commonsense-based shortcuts might be feasible.

Part ID Bias. We also observe no change in performance on TORD instead of an expected decline. We hypothesized that the counterintuitive observation was due to a bias in the questions towards the integer values of part IDs corresponding to the correct answer of TORD questions. To verify this, we try 3 separate runs where we shuffled the part IDs for each part with different seeds, and report the average performance. As shown in Tab. 4, shuffling the part IDs led to a decline in performance on TORD, suggesting that the order of Part IDs was one of the short-cuts that the LVLM was exploiting.

Overall, these results suggest that LVLMs are not using the videos effectively.

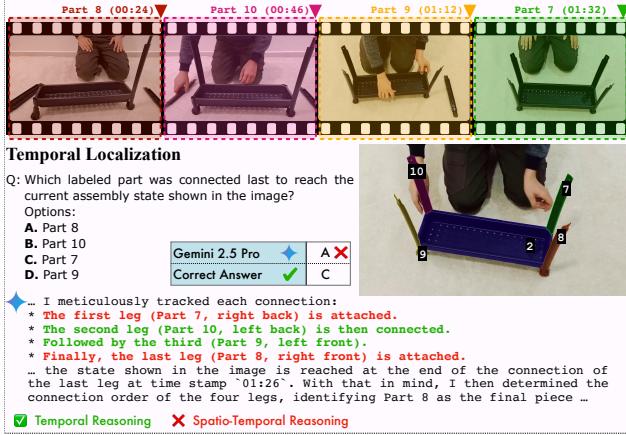


Figure 4. **Self-probing Explanations.** Qualitative example from Gemini 2.5 Pro. We highlight the video with the relevant connection events for clarity. We can observe that the model looks at the video, but makes an error due to gaps in its spatio-temporal reasoning.

5.4. Probing Errors with Self-Explanations

Since LVLMS are failing to effectively utilize the spatio-temporal context in videos, we perform a deeper investigation into such mistakes. To investigate this, we examined the explanations produced by the models while solving our tasks. However, in preliminary experiments, we found that the rationales generated by the open LVLMS were largely paraphrases of its chosen answer, offering little insight into its reasoning. By contrast, the internal thought summaries [6] produced by Gemini 2.5 Pro were substantially richer—featuring time-stamped reasoning and explicit mentions of connection events—which made it far easier to pinpoint the causes of its errors. We therefore base our analysis on these summaries.

Qualitative Example. Figure 4 shows a shortened self-explanation from Gemini 2.5 Pro. Human annotators answered this question correctly. The model demonstrates some temporal localization ability, correctly identifying the timestamp of the final connection event but fails to constrain its subsequent reasoning to that point. It also infers the order of Parts 10 and 9 correctly. However, it mis-tracks the first part to be connected—predicting it as the right back leg (Part 7) instead of the right front leg (Part 8)—inter-changing their position, leading to a wrong answer. For more examples see our project site: flat-pack-bench.github.io/self-explanations.

Error Types from Rationales. In order to do a quantitative analysis on the error types based on the self-explanations, we selected 200 questions where the model got the answer wrong, sampled equally from all 4 question categories. We asked human annotators to write 1-2 sentences on why they think the model committed an error based on the thought summary. Then, for all the questions we collated the question text, the model’s thoughts, the annotators’ comments, asked

an LLM (Gemini 2.5 Pro) to identify 5 error categories that cover the annotations, and assign each question to one or more categories. In this way, we arrived at the following categories (% reflects share within the selected questions):

- **Object Grounding** (37.28%): Failure to correctly identify an object across the image and video.
- **Spatio-Temporal Reasoning** (32.45%): Error in tracking an object’s identity through spatial transformations like camera movement, object rotation, or scene cuts.
- **Temporal Reasoning** (17.98%): The model gets the chronological sequence of object interactions wrong.
- **Physical Interaction** (7.89%): The model misjudges contact, support, or other physical interactions between parts.
- **Language & Logic** (4.38%): Misinterpreting instructions or drawing faulty conclusions from correct observations.

Object grounding and spatio-temporal reasoning are the major error sources, suggesting that the model struggles to understand the fine-grained regional references from the visual prompt and tracking the references through the video.

5.5. Can Task Decomposition Improve Spatio-Temporal Reasoning?

Results so far show that our task is very difficult for current LVLMS to solve in a zero-shot manner. However, we find that most of the tasks in our benchmark can be solved by the use of two primitives: tracking objects (TRACK) and contact reasoning (whether two parts are connected in a particular frame). For instance, consider a Temporal Ordering (TORD) question shown in Fig. 2, where we want to determine the sequence in which parts are connected during assembly. Suppose we have an oracle that provides accurate segmentation maps for each relevant part at every timestamp in the assembly video (the tracking problem), along with their connection status (the contact reasoning problem). We can then iterate through all frames, checking the presence and connection status of each part from the visual prompt. By recording the timestamps at which each part becomes connected and sorting them chronologically, we can effectively answer this question. This raises an important question: Can we design vision systems that can solve the tasks in our benchmark by decomposing them into tracking and contact-reasoning?

TVA: An Agentic Baseline

To test this, we propose a baseline, **Temporal Video Agent (TVA)**, a visual programming agent [25] that has access to tracking and contact-reasoning tools for solving assembly-based tasks as proposed in our benchmark. Following ViperGPT [25], we provide a pythonic API to a *Code LLM* (Gemini 2.5 Pro [7]), that also receives the question as input and generates a program that makes use of the provided API to write a function that can potentially produce an answer. The API includes a video object segmentation function built on top of SAM2 [23], and a VLM-query function (supported by Qwen2.5-VL-32B) that can ask a question related to an

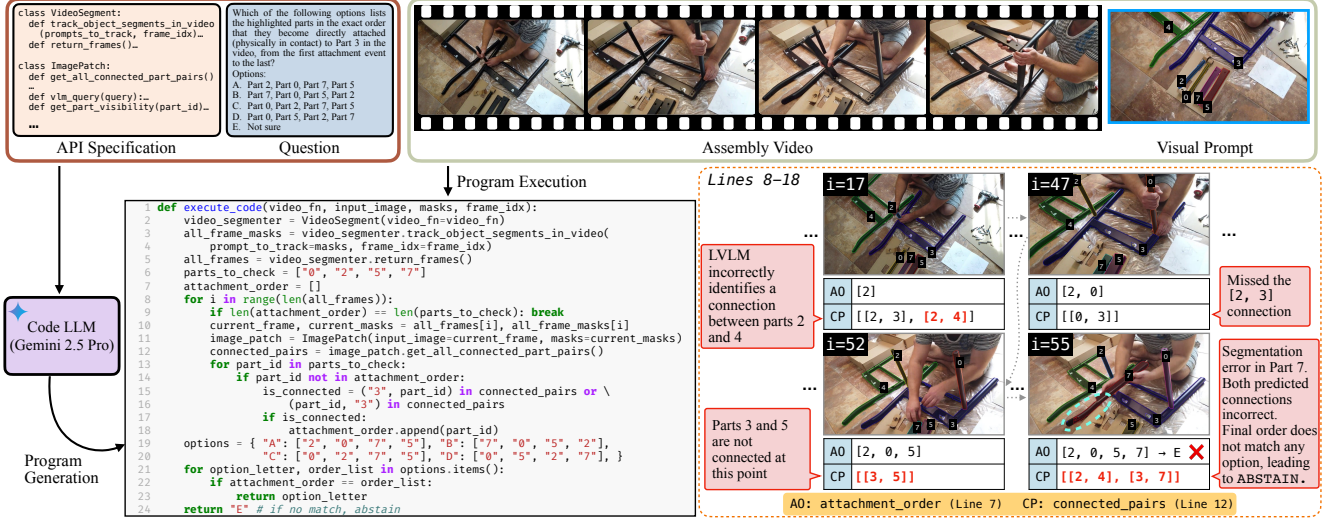


Figure 5. **Temporal Video Agent.** An overview of our agentic baseline. First, a Code LLM uses the API specification and the input question to generate a program. The generated program uses the assembly video and the visual prompt’s frame index and mask to produce a response for the question. We also show an example trace for a question. We can analyse the execution trace to pin-point the sources of error.

image (for resolving contact-reasoning queries) among other tools. While all our questions have a valid answer, it is possible for TVA to end up with a valid program that produces an answer outside of the given options due to tool failures. Hence, we add an ABSTAIN option (“Not Sure”) to each question. We also record the success rates of program execution for the code generated by the orchestrator LLM (See Tab. 5).

Overall TVA performs quite poorly. However, we did find that it is able to correctly answer 11.48% of the questions missed by the LVLM. Figure 5 overviews the method and shows a qualitative example of a generated program and its execution². We observed that inaccuracies in the available tools were the primary causes of errors. We further analysed the performance of individual tools to concretely study the extent of the problem.

Contact-Reasoning Issues. For each pair of visible parts in the visual prompt, we prompted the LVLMs with a binary Yes/No question that asks the models if the two parts are connected as they would be in the final assembly. The ground-truth labels for these questions are mined from the key-frame annotations in IMaW [15]. In this manner, we obtain 1500 questions (750 for each Yes and No). We used Qwen2.5-VL-32B for this evaluation as that was LVLM used in TVA. Our results confirmed what the TVA results hinted: while the LVLM achieves 64.33% accuracy on this task, the

Table 5. **Agent performance.** Acc. (Answered) is accuracy over non-abstained questions

Metric	Value (%)
Accuracy	11.79
Abstained	62.29
Acc. (Answered)	31.27
Program failed	3.32

accuracy on the Yes questions is only **52.93%**, only slightly better than random chance (For more details, see App. D.2).

Tracking Issues. We use the annotated ground-truth segmentations for selected frames in the videos to compute the accuracy of SAM2. We prompt SAM2 [23] with our manually curated segmentation masks in one frame to track them through the video, evaluating these tracks in a second annotated frame. SAM2 achieves a fairly low average IoU of **0.28** (see App. D.2 for details). This suggests that tracking objects in the wild is also a bottleneck for TVA and by extension, would also prove a challenge for LVLMs that must do such tracking implicitly to understand the videos.

These results suggest that beyond the benchmark questions, even simpler tasks on the furniture assembly domain are a challenge for current state-of-the-art.

6. Conclusion

We proposed FLAT-PACK BENCH, a VidQA benchmark that tests LVLMs’ performance on spatio-temporal understanding. Our analysis reveals key bottlenecks of LVLMs in fine-grained video understanding particularly spatio-temporal reasoning (e.g. keeping track of parts through occlusions, scene cuts, etc.) and region-specific grounding. We also study whether an agentic decomposition of the task can help, but find that current vision tools share similar limitations: even specialized tracking models struggle, and LVLMs fail on simpler subproblems such as contact reasoning. Future avenues of work include exploring task-specific fine-tuning on synthetic simulated data, improving visual prompting techniques for regional understanding, and more sophisticated agentic pipelines that can leverage low-level signals like 3D geometry, depth, etc. for improved performance.

²More examples at flat-pack-bench.github.io/tva/

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References

- [1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. [arXiv:2502.13923 \[cs\]](#). 1, 4, 5, 6, 2
- [2] Yuwei Bao, Keunwoo Yu, Yichi Zhang, Shane Storks, Itamar Bar-Yossef, Alex de la Iglesia, Megan Su, Xiao Zheng, and Joyce Chai. Can foundation models watch, talk and guide you step by step to make a cake? In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 12325–12341, Singapore, 2023. Association for Computational Linguistics. 2
- [3] Apratim Bhattacharyya, Bicheng Xu, Sanjay Haresh, Reza Pourreza, Litian Liu, Sunny Panchal, Leonid Sigal, and Roland Memisevic. Can multi-modal LLMs provide live step-by-step task guidance? In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. 2
- [4] Fabian Caba Heilbron, Victor Escorcia, Bernard Ghanem, and Juan Carlos Nibbles. Activitynet: A large-scale video benchmark for human activity understanding. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 961–970, 2015. 1, 2
- [5] Jang Hyun Cho, Andrea Madotto, Effrosyni Mavroudi, Triantafyllos Afouras, Tushar Nagarajan, Muhammad Maaz, Yale Song, Tengyu Ma, Shuming Hu, Hanoona Rasheed, Peize Sun, Po-Yao Huang, Daniel Bolya, Suyog Jain, Miguel Martin, Huiyu Wang, Nikhila Ravi, Shashank Jain, Temmy Stark, Shane Moon, Babak Damavandi, Vivian Lee, Andrew Westbury, Salman Khan, Philipp Krähenbühl, Piotr Dollár, Lorenzo Torresani, Kristen Grauman, and Christoph Feichtenhofer. Perceptionlm: Open-access data and models for detailed visual understanding. [arXiv:2504.13180](#), 2025. 1, 2, 4, 5
- [6] Google AI for Developers. Gemini Thinking — Gemini API Documentation. <https://ai.google.dev/gemini-api/docs/thinking#summaries>, 2025. Accessed: 2025-11-13. 7
- [7] Google DeepMind. Gemini 2.5 pro model card. <https://modelcards.withgoogle.com/assets/documents/gemini-2.5-pro.pdf>, 2025. Accessed: 2025-11-10. 4, 7
- [8] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The” something something” video database for learning and evaluating visual common sense. In *Proceedings of the IEEE international conference on computer vision*, pages 5842–5850, 2017. 2
- [9] Takeshi Kojima, Shixiang (Shane) Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems*, pages 22199–22213. Curran Associates, Inc., 2022. 5
- [10] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 1, 4
- [11] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, Limin Wang, and Yu Qiao. MVBench: A Comprehensive Multi-modal Video Understanding Benchmark, 2024. [arXiv:2311.17005 \[cs\]](#). 1, 2
- [12] Yun Li, Yiming Zhang, Tao Lin, Xiangrui Liu, Wenxiao Cai, Zheng Liu, and Bo Zhao. STI-Bench: Are MLLMs Ready for Precise Spatial-Temporal World Understanding?, 2025. [arXiv:2503.23765 \[cs\]](#). 2
- [13] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023. 4, 2
- [14] Benlin Liu, Yuhao Dong, Yiqin Wang, Zixian Ma, Yansong Tang, Luming Tang, Yongming Rao, Wei-Chiu Ma, and Ranjay Krishna. Coarse correspondences boost spatial-temporal reasoning in multimodal language model. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 3783–3792, 2025. 6
- [15] Yunong Liu, Cristobal Eyzaguirre, Manling Li, Shubh Khanna, Juan Carlos Nibbles, Vineeth Ravi, Saumitra Mishra, Weiyu Liu, and Jiajun Wu. IKEA manuals at work: 4d grounding of assembly instructions on internet videos. In *NeurIPS Datasets and Benchmarks Track*, 2024. 2, 3, 8, 1
- [16] Zuyan Liu, Yuhao Dong, Ziwei Liu, Winston Hu, Jiwen Lu, and Yongming Rao. Oryx MLLM: On-Demand Spatial-Temporal Understanding at Arbitrary Resolution, 2025. [arXiv:2409.12961 \[cs\]](#). 2
- [17] Kartikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023. 1, 2
- [18] Shehan Munasinghe, Hanan Gani, Wenqi Zhu, Jiale Cao, Eric Xing, Fahad Shahbaz Khan, and Salman Khan.

- VideoGLaMM : A Large Multimodal Model for Pixel-Level Visual Grounding in Videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19036–19046, 2025. 2
- [19] OpenAI. Gpt-5 system card. <https://cdn.openai.com/gpt-5-system-card.pdf>, 2025. Accessed: 2025-10-19. 1, 2, 4, 3
- [20] Sunny Panchal, Apratim Bhattacharyya, Guillaume Berger, Antoine Mercier, Cornelius Böhm, Florian Dietrichkeit, Reza Pourreza, Xuanlin Li, Pulkit Madan, Mingu Lee, Mark Todorovich, Ingo Bax, and Roland Memisevic. What to say and when to say it: Live fitness coaching as a testbed for situated interaction. In *Advances in Neural Information Processing Systems*, pages 75853–75882. Curran Associates, Inc., 2024. 2
- [21] Prolific. Prolific. <https://www.prolific.com>, 2025. London, UK. Version used: March 2026. Accessed: 2026-03-25. 3
- [22] Qwen Team. Qwen3-vl github repository. <https://github.com/QwenLM/Qwen3-VL>, 2025. Accessed: 2025-10-20. 4, 5, 2
- [23] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollar, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *The Thirteenth International Conference on Learning Representations*, 2025. 2, 7, 8, 5
- [24] Ye Sun, Hao Zhang, Henghui Ding, Tiehua Zhang, Xingjun Ma, and Yu-Gang Jiang. SAMA: Towards Multi-Turn Referential Grounded Video Chat with Large Language Models. *arXiv preprint arXiv:2505.18812*, 2025. 2
- [25] Dídac Surís, Sachit Menon, and Carl Vondrick. Vipergpt: Visual inference via python execution for reasoning. *Proceedings of IEEE International Conference on Computer Vision (ICCV)*, 2023. 2, 5, 7
- [26] Kexian Tang, Junyao Gao, Yanhong Zeng, Haodong Duan, Yanan Sun, Zhening Xing, Wenran Liu, Kaifeng Lyu, and Kai Chen. Lego-puzzles: How good are mllms at multi-step spatial reasoning? *arXiv preprint arXiv:2503.19990*, 2025. 2
- [27] Zongheng Tang, Yue Liao, Si Liu, Guanbin Li, Xiaojie Jin, Hongxu Jiang, Qian Yu, and Dong Xu. Human-centric spatio-temporal video grounding with visual transformers. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(12):8238–8249, 2021. 2
- [28] Gemini Team et al. Gemini: A Family of Highly Capable Multimodal Models, 2025. *arXiv:2312.11805 [cs]*. 1, 4, 2
- [29] Huaxiaoyue Wang, Gonzalo Gonzalez-Pumariega, Yash Sharma, and Sanjiban Choudhury. Demo2code: From summarizing demonstrations to synthesizing code via extended chain-of-thought. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. 5
- [30] Xin Wang, Taein Kwon, Mahdi Rad, Bowen Pan, Ishani Chakraborty, Sean Andrist, Dan Bohus, Ashley Feniello, Bugra Tekin, Felipe Vieira Frujeri, et al. Holoassist: an egocentric human interaction dataset for interactive ai assistants in the real world. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 20270–20281, 2023. 2
- [31] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023. 5, 7
- [32] Xingrui Wang, Wufei Ma, Angtian Wang, Shuo Chen, Adam Kortylewski, and Alan Yuille. Compositional 4D Dynamic Scenes Understanding with Physics Priors for Video Question Answering, 2025. *arXiv:2406.00622 [cs]*. 1, 2
- [33] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. In *NeurIPS*, 2024. 2
- [34] Junbin Xiao, Xindi Shang, Angela Yao, and Tat-Seng Chua. NExT-QA: Next Phase of Question-Answering to Explaining Temporal Actions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9777–9786, 2021. 1
- [35] Zihui Xue, Mi Luo, and Kristen Grauman. Seeing the arrow of time in large multimodal models. *arXiv preprint arXiv:2506.03340*, 2025. 2, 4, 5
- [36] Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-mark prompting unleashes extraordinary visual grounding in gpt-4v. *arXiv preprint arXiv:2310.11441*, 2023. 3
- [37] Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in Space: How Multimodal Large Language Models See, Remember and Recall Spaces. *arXiv preprint arXiv:2412.14171*, 2024. 2, 4, 7
- [38] Linli Yao, Haoning Wu, Kun Ouyang, Yuanxing Zhang, Caiming Xiong, Bei Chen, Xu Sun, and Junnan Li. Generative frame sampler for long video understanding. In *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria, 2025. Association for Computational Linguistics. 4, 3
- [39] Yuqian Yuan, Hang Zhang, Wentong Li, Zesen Cheng, Boqiang Zhang, Long Li, Xin Li, Deli Zhao, Wenqiao Zhang, Yueting Zhuang, Jianke Zhu, and Lidong Bing. VideoRefer Suite: Advancing Spatial-Temporal Object Understanding with Video LLM, 2025. *arXiv:2501.00599 [cs]*. 1, 2, 4, 5, 3
- [40] Weichen Zhang, Ruiying Peng, Chen Gao, Jianjie Fang, Xin Zeng, Kaiyuan Li, Ziyu Wang, Jinqiang Cui, Xin Wang, Xinlei Chen, and Yong Li. The Point, the Vision and the Text: Does Point Cloud Boost Spatial Reasoning of Large Language Models?, 2025. *arXiv:2504.04540 [cs]*. 2
- [41] Yuanhan Zhang, Bo Li, haotian Liu, Yong jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model. <https://llava-vl.github.io/blog/2024-04-30-llava-next-video/>, 2024. Accessed March 15, 2026. 1, 4
- [42] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. LLaVA-Video: Video Instruction Tuning With Synthetic Data, 2025. *arXiv:2410.02713 [cs]*. 1, 4, 3

- [43] Zhu Zhang, Zhou Zhao, Yang Zhao, Qi Wang, Huasheng Liu, and Lianli Gao. Where does it exist: Spatio-temporal video grounding for multi-form sentences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10668–10677, 2020. [2](#)
- [44] Zhicheng Zheng, Xin Yan, Zhenfang Chen, Jingzhou Wang, Qin Zhi Eddie Lim, Joshua B Tenenbaum, and Chuang Gan. Contphy: Continuum physical concept learning and reasoning from videos. In *International Conference on Machine Learning*. PMLR, 2024. [1](#), [2](#)
- [45] Baichuan Zhou, Haote Yang, Dairong Chen, Junyan Ye, Tianyi Bai, Jinhua Yu, Songyang Zhang, Dahua Lin, Conghui He, and Weijia Li. UrBench: A Comprehensive Benchmark for Evaluating Large Multimodal Models in Multi-View Urban Scenarios, 2025. arXiv:2408.17267 [cs]. [2](#)
- [46] Honglu Zhou, Xiangyu Peng, Shrikant Kendre, Michael S. Ryoo, Silvio Savarese, Caiming Xiong, and Juan Carlos Niebles. Strefer: Empowering Video LLMs with Space-Time Referring and Reasoning via Synthetic Instruction Data, 2025. arXiv:2509.03501 [cs]. [1](#), [2](#)
- [47] Shijie Zhou, Alexander Vilesov, Xuehai He, Ziyu Wan, Shuwang Zhang, Aditya Nagachandra, Di Chang, Dongdong Chen, Eric Xin Wang, and Achuta Kadambi. Vlm4d: Towards spatiotemporal awareness in vision language models. In *Proceedings of the IEEE/CVF international conference on computer vision*, 2025. [2](#)
- [48] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Xuehui Wang, Yue Cao, Yangzhou Liu, Xingguang Wei, Hongjie Zhang, Haomin Wang, Weiye Xu, Hao Li, Jiahao Wang, Nianchen Deng, Songze Li, Yinan He, Tan Jiang, Jiapeng Luo, Yi Wang, Conghui He, Botian Shi, Xingcheng Zhang, Wenqi Shao, Junjun He, Yingtong Xiong, Wenwen Qu, Peng Sun, Penglong Jiao, Han Lv, Lijun Wu, Kaipeng Zhang, Huipeng Deng, Jiaye Ge, Kai Chen, Limin Wang, Min Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai, and Wenhui Wang. InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models, 2025. arXiv:2504.10479 [cs]. [1](#), [4](#), [5](#), [7](#)
- [49] Orr Zohar, Xiaohan Wang, Yann Dubois, Nikhil Mehta, Tong Xiao, Philippe Hansen-Estruch, Licheng Yu, Xiaofang Wang, Felix Juefei-Xu, Ning Zhang, Serena Yeung-Levy, and Xide Xia. Apollo: An Exploration of Video Understanding in Large Multimodal Models, 2024. arXiv:2412.10360 [cs]. [2](#)